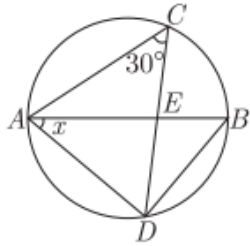


Chapter

Angle Properties Of Circle

1. In the given circle with diameter AB, find the value of x .

Comp 2023

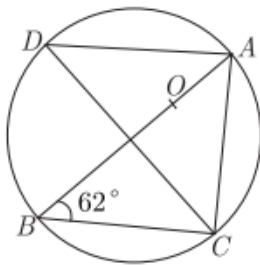


2. In the given figure A, B, C and D are points on the circle with centre O. Given $\angle ABC = 62^\circ$. Find :

(i) $\angle ADC$

(ii) $\angle BAC$

Main 2022 Sem II

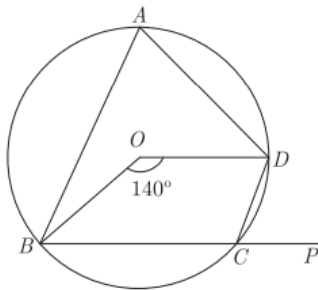


3. In given figure, O is the centre of the circle. The angle subtended by the arc BCD at the centre is 140° . BC is produced to P.

(i) Find $\angle BAD$

(ii) Find $\angle DCB$

Comp 2022



Solution

1. $\angle ABD = \angle ACD = 30^\circ$...[Angle in the same segment]

Now in $\triangle ADB$,

$\angle BAD + \angle ADB + \angle ABD = 180^\circ$...[Angles of a triangle]

But, $\angle ADB = 90^\circ$...[Angle in a semi-circle]

$$\therefore x + 90^\circ + 30^\circ = 180^\circ$$

$$\Rightarrow x + 120^\circ = 180^\circ$$

$$\Rightarrow x = 180^\circ - 120^\circ$$

$$\Rightarrow x = 60^\circ$$

2. Given:

$$\angle ABC = 62^\circ$$

(i) Find $\angle ADC$

Angles in the **same segment** of a circle are equal.

$$\angle ADC = \angle ABC = 62^\circ$$

(ii) Find $\angle BAC$

$\angle ABC$ subtends arc AC , so the **central angle**:

$$\angle AOC = 2 \times 62^\circ = 124^\circ$$

In $\triangle AOC$ ($OA = OC$),

$$\angle BAC = \frac{180^\circ - 124^\circ}{2} = \boxed{28^\circ}$$

Final Answers:

- (i) $\angle ADC = 62^\circ$
- (ii) $\angle BAC = 28^\circ$

3. Given:

Angle subtended by arc **BCD** at the centre $O = 140^\circ$ $\angle O = 140^\circ$

So, arc **BCD** = 140°

(i) Find $\angle BAD$

$\angle BAD$ is an **inscribed angle** subtending the same arc **BCD**.

$$\angle BAD = \frac{1}{2} \times 140^\circ = \boxed{70^\circ}$$

(ii) Find $\angle DCB$

$\angle DCB$ subtends the **major arc BD**.

Major arc $BD = 360^\circ - 140^\circ = 220^\circ$

$$\angle DCB = \frac{1}{2} \times 220^\circ = \boxed{110^\circ}$$

Final Answers:

- (i) $\angle BAD = 70^\circ$
- (ii) $\angle DCB = 110^\circ$